Notes A Functorial Excursion Between Algebraic Geometry and Linear Logic

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Contents

1 Basics **1**

2 On the category of rings **2**

3 On the category of R-modules and modules **3**

4 The category $\mathbf{Mod}^{\ominus}$ of modules and retromorphisms **5**

5 The category $\mathbf{Mod}_R$ is symmetric monoidal closed **6**

1 Basics

Let us start by recalling basic principles of algebraic geometry. The intuition is that a space X comes along with a topological space (X, Σ_X) such that every open set U is equipped with a set $\mathcal{O}_X(U)$ of local operations called *regular functions*. Such regular functions are typically defined as polynomials with division and each $\mathcal{O}_X(U)$ defines a commutative ring. In functorial terms, every space gives rise to a ringed space, that is, a topological space (X, Σ_X) equipped with a contravariant functor $\mathcal{O}_X : \Sigma_X^{op} \to \mathbf{Ring}$ from the set of opens Σ_X^{op} with the reversed inclusion order. The functorial action of $\mathcal{O}_X$ describes the restriction functions

$$\mathcal{O}_X(V \subseteq U) : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$$

which takes an operation $f \in \mathcal{O}_X(U)$ defined on an open set U and restrict it to an operation

$$f|_V := \mathcal{O}_X(V \subseteq U)(f)$$

on an open subset $V \subseteq U$. The functorial understanding of the concept of a space is supported by the observation that every commutative ring R defines a ringed space:

$$\operatorname{Spec} R = (\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$$

on the set of prime ideals $\mathfrak{p}$ of the commutative ring R. Such a ringed space is called the *affine scheme* associated to R. Every space X of the theory, called a *scheme*, is a patchwork of affine schemes $\operatorname{Spec} R_i$ glued together using the language of sheaf theory.

Every scheme X comes equipped with a symmetric monoidal category $\mathbf{qcMod}_X$ of quasicoherent sheaf of $\mathcal{O}_X$-modules. By a presheaf of $\mathcal{O}_X$-modules M of a ringed space $(X, \mathcal{O}_X)$, we mean a contravariant functor

$$M : \Omega_X^{op} \to \mathbf{Ab}$$

where $\mathbf{Ab}$ is the category of Abelian groups such that each Abelian group $M(U)$ defines a $\mathcal{O}_X(U)$-module. The Serre-Swann theorem states a vector bundle on a suitable ringed space $(X, \mathcal{O}_X)$ can be equivalently encoded as its sheaf of sections that defines a locally free and projective sheaf of $\mathcal{O}_X$-modules. A sheaf of $\mathcal{O}_X$ is called quasicoherent when it is locally presentable, that is, it is locally the cokernel of a morphism of free modules. The category $\mathbf{qcMod}_X$ extends the category of vector bundles to define an Abelian category where kernels and direct images can be computed.

The idea is to associate a model of intuitionistic linear logic, where formulas are understood as generalised vector bundles and proofs are vector fields by building an appropriate linear-non-linear adjunction. To associate a model of linear logic with each scheme X, we take the category $\mathbf{PshMod}_X$ of presheafs of $\mathcal{O}_X$-modules and their homomorphisms as the underlying symmetric monoidal closed category with tensor product $\otimes_X$, the structural sheaf $\mathcal{O}_X$ as unit, and $\multimap_X$ for internal homs. For the cartesian part of a linear-non-linear adjunction, we take the category $\mathbf{PshCoAlg}_X$ of cocommutative comonoids (or commutative coalgebras) in $\mathbf{PshMod}_X$. It is convenient to the approach based on the functor of points (see [EH00, I.4]). Spec defines a contravariant functor $\operatorname{Spec} : \mathbf{Ring}^{op} \to \mathbf{Scheme}$ that induces the *nerve functor*

$$\mathbf{nerve} : \mathbf{Scheme} \to [\mathbf{Ring}, \mathbf{Set}]$$

associating to every commutative ring R *the set of R-points in a scheme X* defined as $\mathbf{Scheme}(\operatorname{Spec} R, X)$. The nerve functor is fully faithful and it allows one to see every scheme X as a covariant presheaf

$$\mathbf{nerve}(X) : \mathbf{Ring} \to \mathbf{Set}$$

So we move to schemes to $\mathbf{Ring}$-spaces and associate a model of intuitionistic linear logic with each $\mathbf{Ring}$-space X, where formulas are interpreted in $\mathbf{PshMod}_X$ as presheaves of $\mathcal{O}_X$-module understood as generalised vector bundles on the $\mathbf{Ring}$-space. The linear-non-linear adjunction (and therefore the exponential modality) is constructed by the Sweedler construction.

2 On the category of rings

Let $(\mathcal{A}, \otimes, \mathbb{1})$ be a symmetric monoidal category with reflexive coequalisers preserved by tensor products in every coordinate. For example, the category $\mathbf{Ab}$ of Abelian groups and homomorphisms. A ring is a monoid object (R, m, e) in

$(\mathcal{A}, \otimes, \mathbb{1})$. In other words, a ring is a triple such that (R, m, e) consisting of an object R and morphisms $m : R \otimes R \to R$ and $e : \mathbb{1} \to R$ such that the following diagrams commute:

$$\begin{tikzcd}
R \otimes R \otimes R \arrow[r, "m \otimes R"] \arrow[d, "R \otimes m"'] & R \otimes R \arrow[d, "m"] \\
R \otimes R \arrow[r, "m"'] & R
\end{tikzcd}
\qquad
\begin{tikzcd}
R \otimes R \arrow[r, "m"] \arrow[d, "m"'] & R \arrow[d, "e \otimes R"] \\
R \arrow[r, "R \otimes e"'] \arrow[ur, equal] & R \otimes R
\end{tikzcd}$$

A ring (R, m, e) is commutative is the following triangle commutes:

$$\begin{tikzcd}
R \otimes R \arrow[rr, "\gamma_{R,R}"] \arrow[dr, "m"'] & & R \otimes R \arrow[dl, "m"] \\
& R &
\end{tikzcd}$$

A *ring morphism* $u : (R, m_R, e_R) \to (S, m_S, e_S)$ is a morphism $u : R \to S$ in the underlying category $\mathcal{A}$ making the following diagrams commute:

$$\begin{tikzcd}
R \otimes R \arrow[r, "u \otimes u"] \arrow[d, "m_S"'] & S \otimes S \arrow[d, "m_S"] \\
R \arrow[r, "u"'] & S
\end{tikzcd}
\qquad
\begin{tikzcd}
& \mathbb{1} \arrow[dl, "e_R"'] \arrow[dr, "e_S"] & \\
R \arrow[rr, "u"'] & & S
\end{tikzcd}$$

The category $\mathbf{Ring}(\mathcal{A})$ of rings in $\mathcal{A}$ defined by the tensor product $R, S \mapsto R \otimes S$ in $\mathcal{A}$.

3 On the category of R-modules and modules

Given a ring R in a symmetric monoidal category $\mathcal{A}$, an *R-module* is a pair (M, act) where M is an object in $\mathcal{A}$ along with a morphism $\mathrm{act} : R \otimes M \to M$ in $\mathcal{A}$ such that:

$$\begin{tikzcd}
R \otimes R \otimes M \arrow[r, "m_R \otimes M"] \arrow[d, "R \otimes \mathrm{act}"'] & R \otimes M \arrow[d, "\mathrm{act}"] \\
R \otimes M \arrow[r, "\mathrm{act}"'] & M
\end{tikzcd}
\qquad
\begin{tikzcd}
& R \otimes M \arrow[dr, "\mathrm{act}"] & \\
M \arrow[ur, "e_R \otimes M"] \arrow[rr, equal] & & M
\end{tikzcd}$$

Equivalently, an R-module is an Eilenberg-Moore algebra for the monad $A \mapsto R \otimes A$. An R-module homomorphism $f : M \to N$ between two modules is a map f in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{tikzcd}
R \otimes M \arrow[r, "R \otimes f"] \arrow[d, "\mathrm{act}_M"'] & R \otimes N \arrow[d, "\mathrm{act}_N"] \\
M \arrow[r, "f"'] & N
\end{tikzcd}$$

$\mathbf{Mod}_R$ is the category of R-modules and their morphisms.

A *module* is a pair (R, M) where R is a ring and M is a module. A *module homomorphism* $(u, f) : (R, M) \to (S, N)$ is a ring morphism $u : R \to S$ and a morphism $f : M \to N$ such that

$$\begin{tikzcd}
R \otimes M \arrow[r, "u\otimes f"] \arrow[d, "\mathrm{act}_M"'] & S \otimes N \arrow[d, "\mathrm{act}_N"] \\
M \arrow[r, "f"'] & N
\end{tikzcd}$$

$\mathbf{Mod}$ is the category of modules and their morphisms.

There is a functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ transporting every module (R, M) to the underlying ring R and every morphism (u, f) to u. We will write

$$u : R \to S \models f : M \to N$$

for a module morphism $(u, f) : (R, M) \to (S, N)$. The intuition behind the notation is that every ring homomorphism $u : R \to S$ induces a "fibre" consisting of all module morphisms $(u, f) : (R, M) \to (S, N)$ living above and refining the ring morphism $u : R \to S$. The fibre of πR for a commutative ring is the category $\mathbf{Mod}_R$. The functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ defines a Grothendieck fibration: every pair consisting of a ring morphism $u : R \to S$ and of an S-module (N, act_N) induces an R-module noted

$$\mathbf{res}_u N = (N, \mathrm{act}'_N)$$

with the action morphism act'_N defined as the composite:

$$R \otimes N \xrightarrow{u\otimes N} S \otimes N \xrightarrow{\mathrm{act}_N} N$$

Moreover, the S-module with a module morphism:

$$u : R \to S \models id_N : \mathbf{res}_u N \to N$$

which is weakly Cartesian in the sense that every module morphism

$$u : R \to S \models f : M \to N$$

factors uniquely as

$$M \xrightarrow{(id_R, h)} \mathbf{res}_u N \xrightarrow{(u, id_N)} N$$

for a morphism $h : M \to \mathbf{res}_u N$ in the category $\mathbf{Mod}_R$. The unique solution h is equal to the original morphism $f : M \to N$ viewed as a morphism in $\mathbf{Mod}_R$. Given a ring morphism $u : R \to S$, the existence of a weakly Cartesian map for every S-module N ensures the existence of a functor

$$\mathbf{res}_u : \mathbf{Mod}_S \to \mathbf{Mod}_R$$

called restriction of scalar along u. Along with the fact that the composite of weakly Cartesian maps is weakly Cartesian we have that the forgetful functor $\pi : \mathbf{Mod} \to \mathbf{Ring}$ is a Grothendieck fibration.

Now we use that fact the category $\mathcal{A}$ has reflexive coequalisers, so one can show that $\mathbf{res}_u$ has a left adjoint

$$\mathbf{ext}_u : \mathbf{Mod}_R \to \mathbf{Mod}_S$$

constructed as follows. Every ring homomorphism $u : R \to S$ induces an $R \otimes S$-module notes as $R \otimes_u S$ defined as the reflexive coequaliser of the following diagram

$$\begin{tikzcd} R \otimes R \otimes S \arrow[r, shift left=2, "m_R \otimes S"] \arrow[r, shift right=2, "(R\otimes m_S)\circ(R\otimes u\otimes S)"'] & R \otimes S \arrow[l, "R\otimes e_R \otimes S"' description] \end{tikzcd}$$

computed in $\mathcal{A}$. Given three commutative rings R, S_1 and S_2, we define the functor

$$\circledast_R : \mathbf{Mod}_{S_1 \otimes R} \times \mathbf{Mod}_{R \otimes S_2} \to \mathbf{Mod}_{S_1 \otimes S_2}$$

The two maps $\mathrm{act}_M : M \otimes R \to M$ and $\mathrm{act}_N : N \otimes R \to N$ are deduced by restriction of scalar from the $S_1 \otimes R$-module structure of M and the $R \otimes S_2$-module structure of N. The left adjunction functor $\mathbf{ext}_u$ is then defined the following way:

$$\mathbf{ext}_u : M \mapsto M \circledast_R (R \otimes_u S)$$

4 The category $\mathbf{Mod}^{\ominus}$ of modules and retromorphisms

Let $[M, N]$ denote the internal hom of M and N in $\mathcal{A}$. A *module retromorphism*

$$(u, f) : (S, N) \to (R, M)$$

is a pair (u, f) where $u : R \to S$ is a ring morphism and $f : N \to M$ is a morphism in $\mathcal{A}$ such that the following diagram commutes:

$$\begin{tikzcd} R \otimes M \arrow[d, "\mathrm{act}_M"'] & R \otimes N \arrow[l, "R\otimes f"'] \arrow[r, "u \otimes N"] & S \otimes N \arrow[d, "\mathrm{act}_N"] \\ M & & N \arrow[ll, "f"] \end{tikzcd}$$

The category $\mathbf{Mod}^{\ominus}$ has ring modules as objects and module retromorphisms. There is a forgetful functor $\pi : \mathbf{Mod}^{\ominus} \to \mathbf{Ring}$ transporting every retromorphism to the underlying ring morphism and π is also a Grothendieck fibration. π is also a bifibration. Let $u : R \to S$ be a ring morphism, then functor defined by coextension of scalar along u

$$\mathbf{coext}_u : \mathbf{Mod}_R \to \mathbf{Mod}_S$$

transport every R-module (M, act_M) to the S-module $[S, M]_u$ defined as the coreflexive equaliser of the below diagram:

$$
\begin{tikzcd}
{[S,M]} & {[R \otimes S, M]} \arrow[l, "{[e_R \otimes M, M]}" description] \arrow[from=1-1, to=1-2, shift left=3, "{[u\otimes S, M]\circ[m_S, M]}"] \arrow[from=1-1, to=1-2, shift right=3, "{[R\otimes S, \mathrm{act}_M]\circ[R\otimes\_, R\otimes\_]}"']
\end{tikzcd}
$$

Note that $[S, M]_u$ provides an internal description of the set of morphisms $f : S \to M$ making the following diagram commute:

$$
\begin{tikzcd}
R \otimes S \arrow[r, "R\otimes f"] \arrow[d, "u\otimes S"'] & R \otimes M \arrow[dd, "\mathrm{act}_M"] \\
S \otimes S \arrow[d, "m_S"'] & \\
S \arrow[r, "f"'] & M
\end{tikzcd}
$$

or, equivalently, as the set of morphisms $f : \mathbf{res}_u S \to M$. Thus, every ring morphism $u : R \to S$ induces the following sequence of adjunctions:

$$
\begin{tikzcd}
\mathbf{Mod}_R \arrow[r, shift left=3, "\mathbf{coext}_u"] \arrow[r, shift right=3, "\mathbf{ext}_u"'] & \mathbf{Mod}_S \arrow[l, "\mathbf{res}_u" description]
\end{tikzcd}
$$

5 The category $\mathbf{Mod}_R$ is symmetric monoidal closed

References

[EH00] David Eisenbud and Joe Harris. *The geometry of schemes*. Springer, 2000.